

Fundamental Calculus/Differential and Integral Calculus

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Reference : Calculus 10-th Edition by Howard Anton,
Irl Bivens and Stephen Davis

Chapter : 0.2(Page-15)

NEW FUNCTIONS FROM OLD

Sketch the graph of $f(x)=y = \frac{1}{x}$, also find their domain and range.

- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = \frac{1}{x} \Rightarrow x = \frac{1}{y}, y \neq 0$, so range is $y \neq 0$

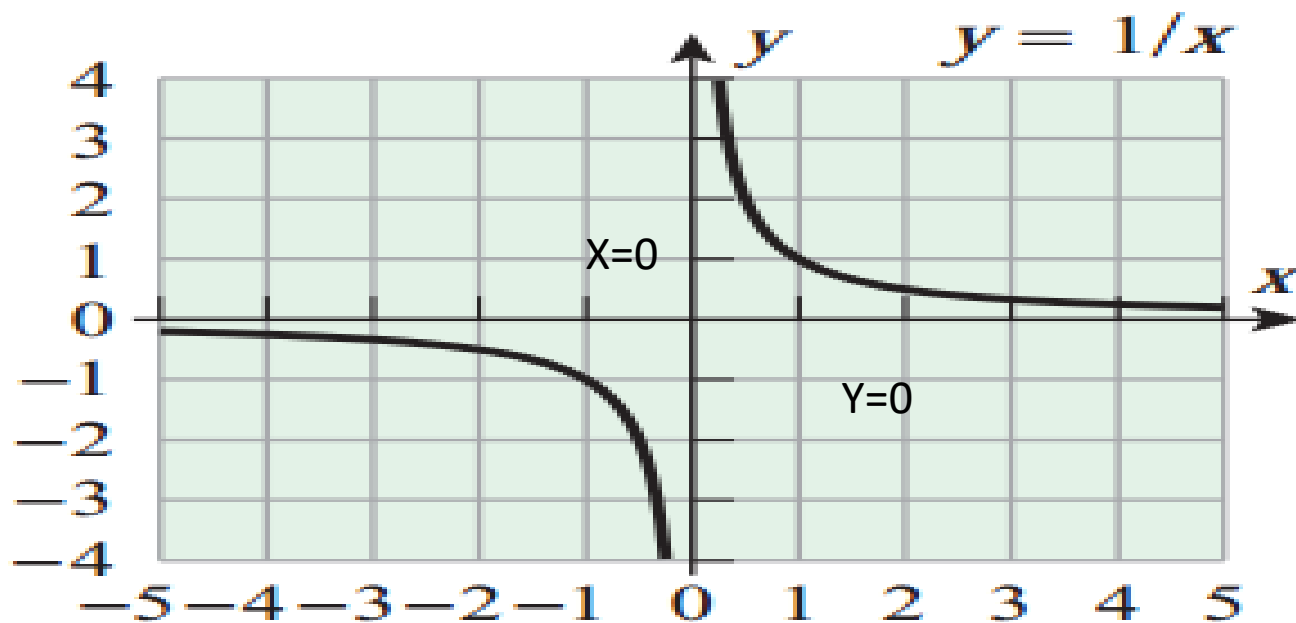
Asymptote line $x = 0, y = 0$,

Or

Domain is : $R - \{0\}$ and Range is : $R - \{0\}$ or ,

Domain is $(-\infty, 0) \cup (0, +\infty)$ and Range is $(-\infty, 0) \cup (0, +\infty)$

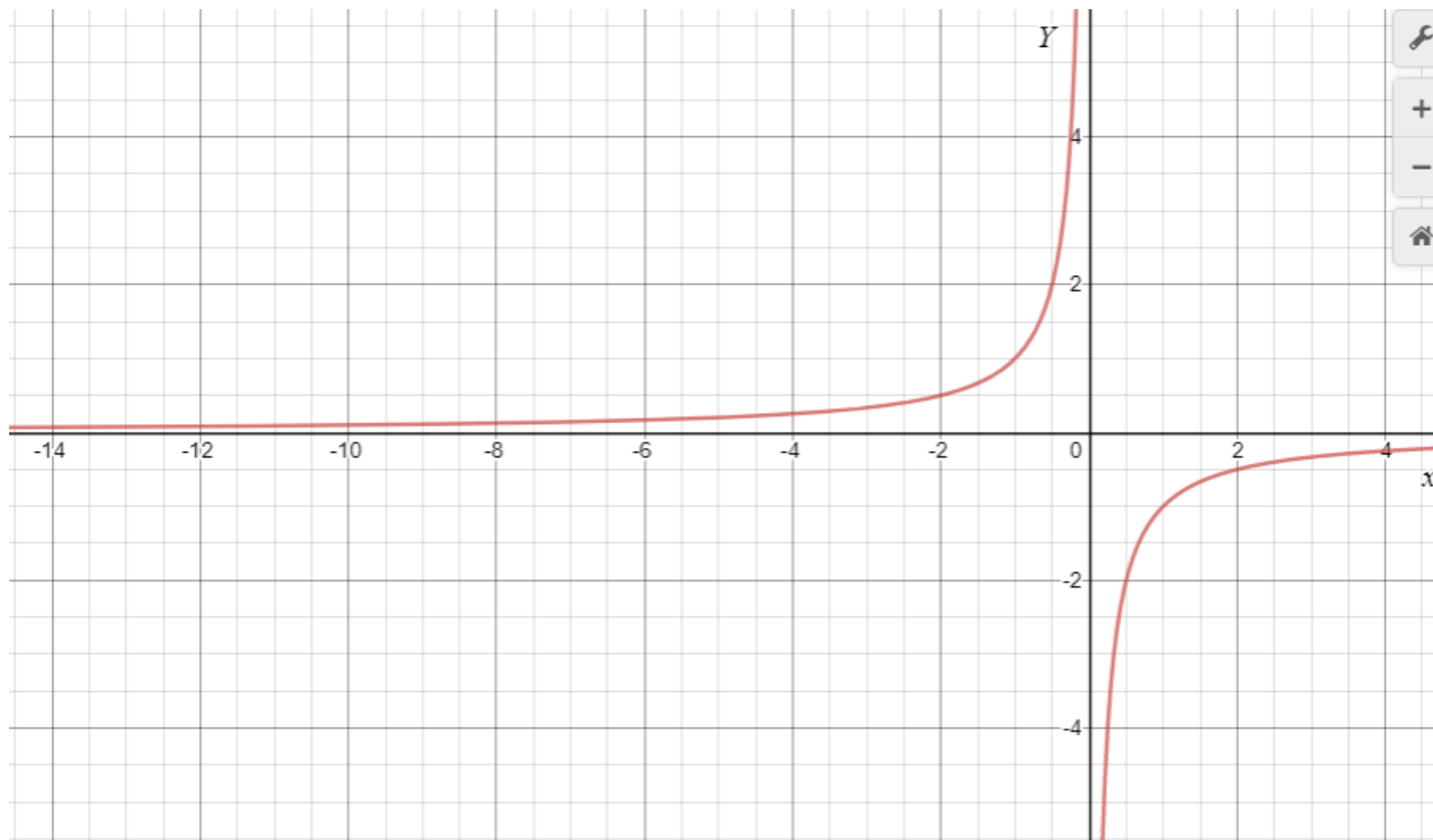
Graph of $f(x)=y = \frac{1}{x}$



Sketch the graph of $f(x)=y = -\frac{1}{x}$, also find their domain and range.

- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = -\frac{1}{x} \Rightarrow x = -\frac{1}{y}$, $y \neq 0$, so range is $y \neq 0$
- Asymptote line $x = 0$, $y = 0$,
- Domain is $\mathbb{R}-\{0\}$ or $x \neq 0$
- Range is $\mathbb{R}-\{0\}$ or $y \neq 0$

Graph of $f(x)=y = -\frac{1}{x}$



Show all

×

Sketch the graph of $f(x)=y = 1 + \frac{1}{x}$, also find their domain and range.

- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = 1 + \frac{1}{x} \Rightarrow y - 1 = \frac{1}{x} \Rightarrow xy - x = 1$
- $\Rightarrow x(y - 1) = 1 \Rightarrow x = \frac{1}{(y-1)}, y \neq 1$

, so range is $y \neq 1$

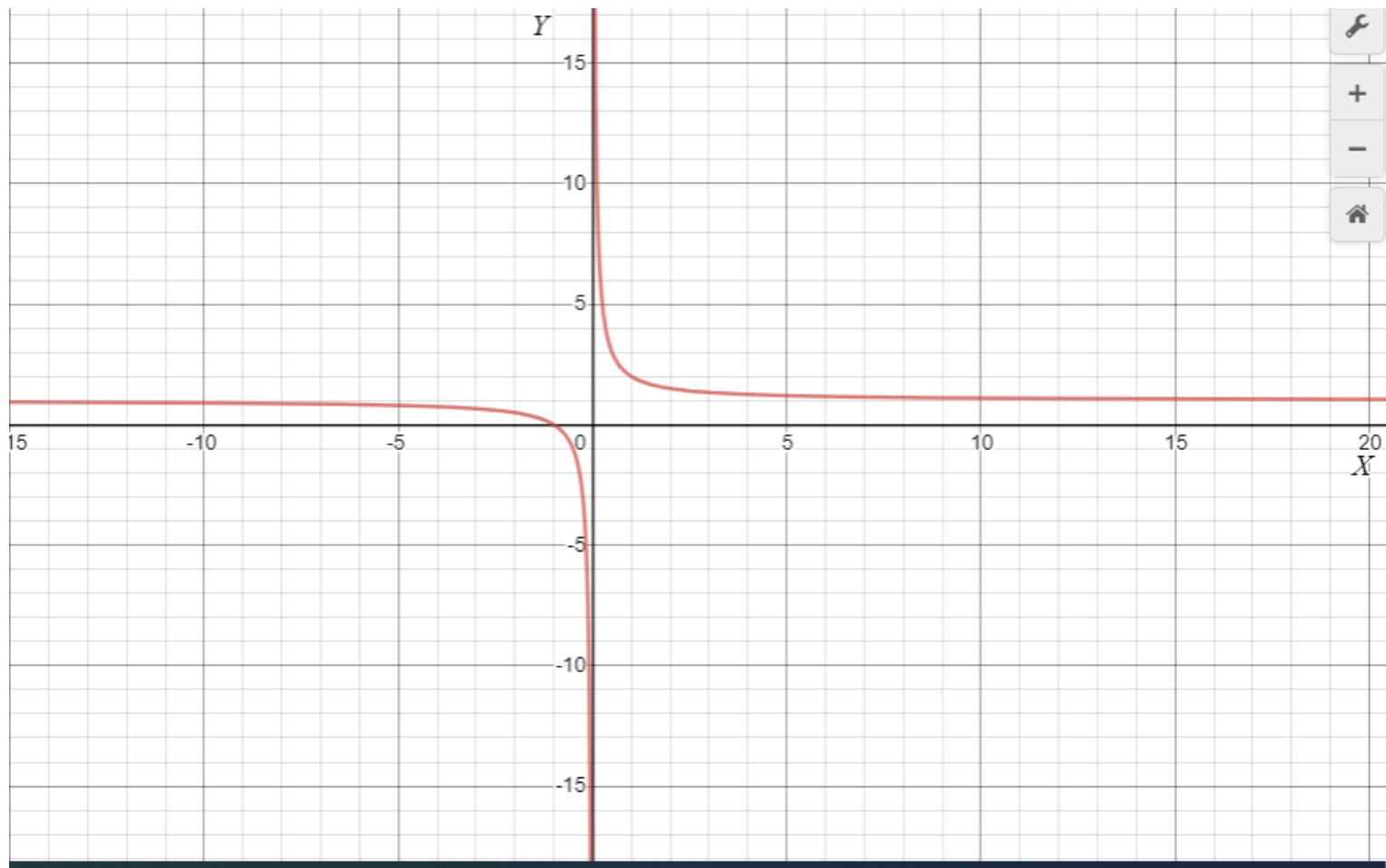
Asymptote line $x = 0, y = 0 + 1 = 1$,

Or

Domain is : $\mathbb{R} - \{0\}$ and Range is : $\mathbb{R} - \{1\}$ or ,

Domain is $(-\infty, 0) \cup (0, +\infty)$ and Range is $(-\infty, 1) \cup (1, +\infty)$

Graph of $f(x)=y = 1 + \frac{1}{x}$



Sketch the graph of $f(x)=y = 1 - \frac{1}{x}$, also find their domain and range.

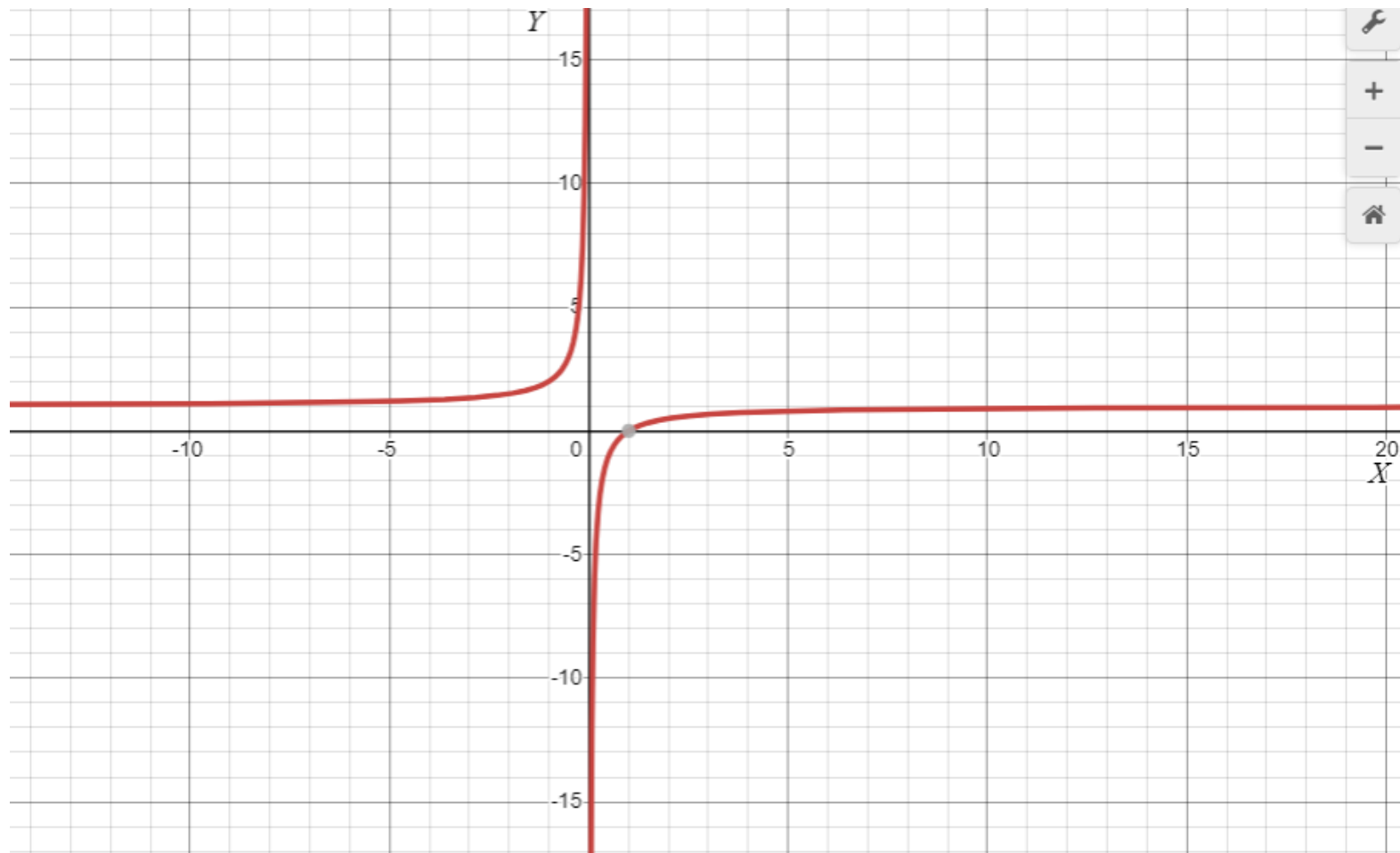
- Since f is undefined if $x = 0$, so domain is $x \neq 0$.
- Here $y = 1 - \frac{1}{x} \Rightarrow -y + 1 = \frac{1}{x} \Rightarrow -xy + x = 1$
- $\Rightarrow x(1 - y) = 1 \Rightarrow x = \frac{1}{1-y}, y \neq 1$
- , , so range is $y \neq 1$

Or

Domain is : $R - \{0\}$ and Range is : $R - \{1\}$ or ,

Domain is $(-\infty, 0) \cup (0, +\infty)$ and Range is $(-\infty, 1) \cup (1, +\infty)$

Graph of $f(x)=y = 1 - \frac{1}{x}$

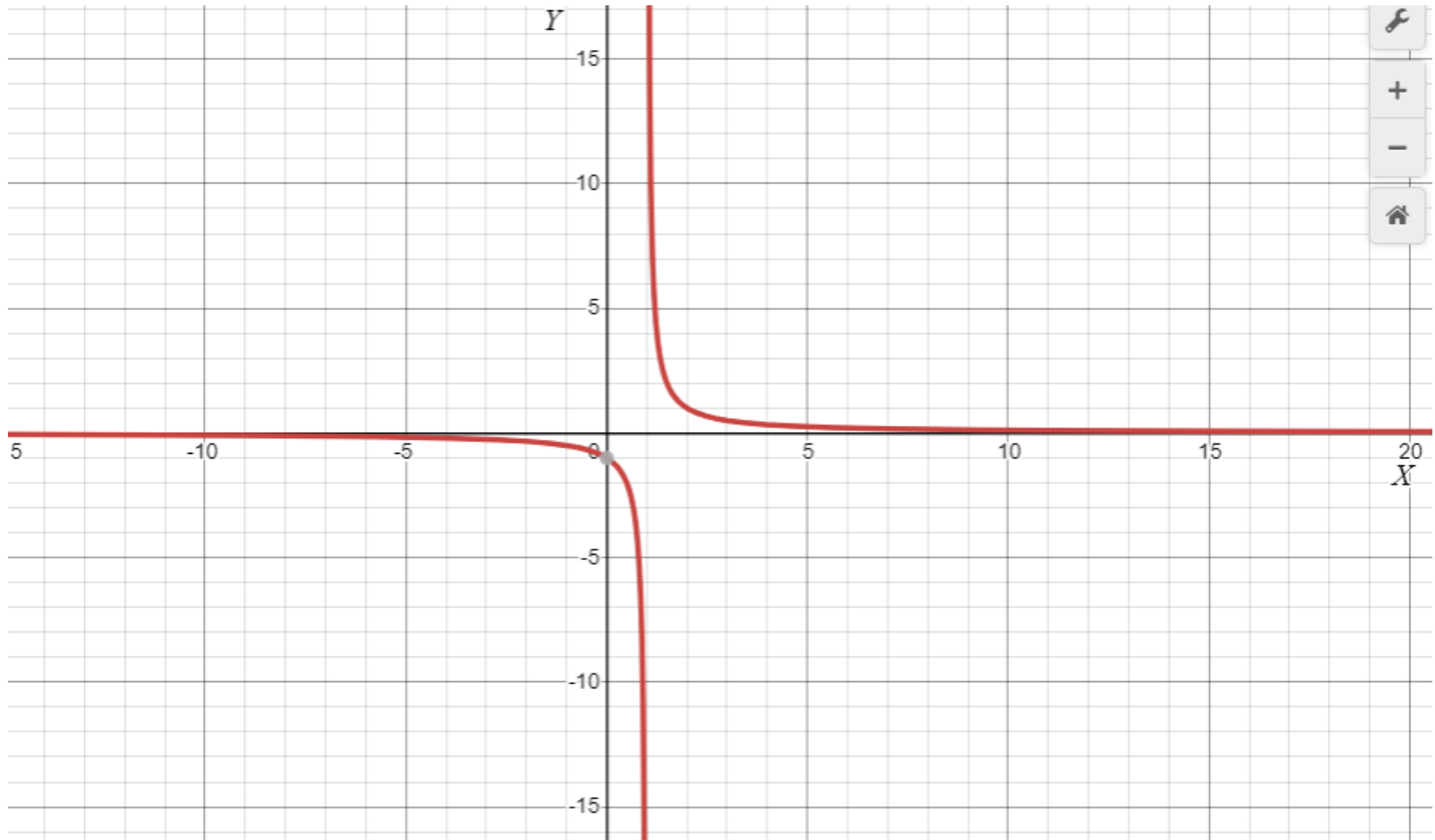


Home work

- $y = \frac{x-1}{x} = \frac{x}{x} - \frac{1}{x} = 1 - \frac{1}{x}$ [Home work]

- $y = \frac{x}{x-1} = \frac{x-1+1}{x-1} = \frac{x-1}{x-1} + \frac{1}{x-1}$
 $= 1 + \frac{1}{x-1}$ [Home work] $x=1, y=1$

Sketch the graph of $f(x)=y = \frac{1}{x-1}$, also find their domain and range.

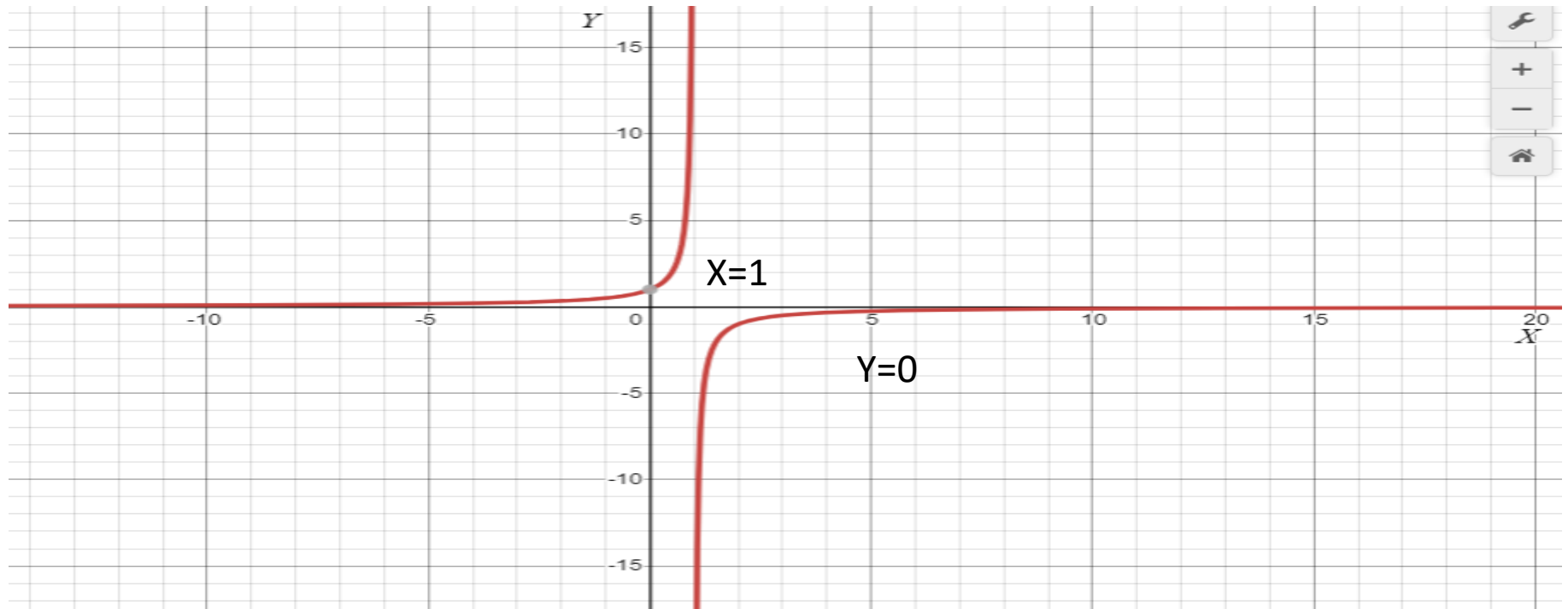


Finding domain and range $y = \frac{1}{x-1}$.

- Since f is undefined if $x - 1 = 0$
 $\Rightarrow x = 1$ so domain is $x \neq 1$.
- Here $y = \frac{1}{x-1} \Rightarrow xy - y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1+y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{1\}$ and Range is : $R - \{0\}$

Sketch the graph of $f(x)=y = \frac{1}{1-x}$,
also find their domain and range.

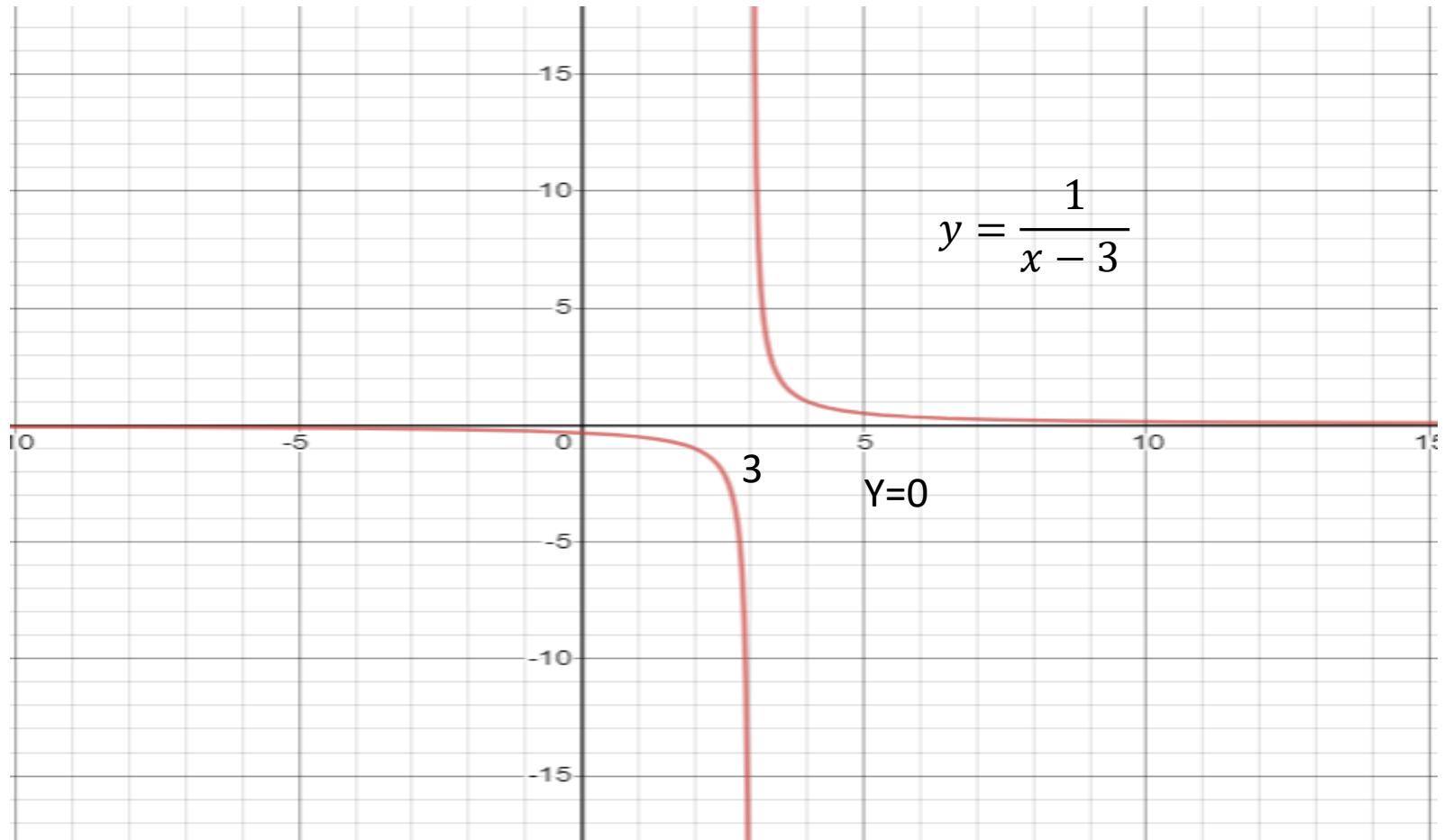
- $y = \frac{1}{1-x} = \frac{1}{-(x-1)}$



Finding domain and range $y = \frac{1}{1-x}$.

- Since f is undefined if $1-x = 0$
 $\Rightarrow x = 1$ so domain is $x \neq 1$.
- Here $y = \frac{1}{1-x} \Rightarrow -xy + y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1-y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{1\}$ and Range is : $R - \{0\}$

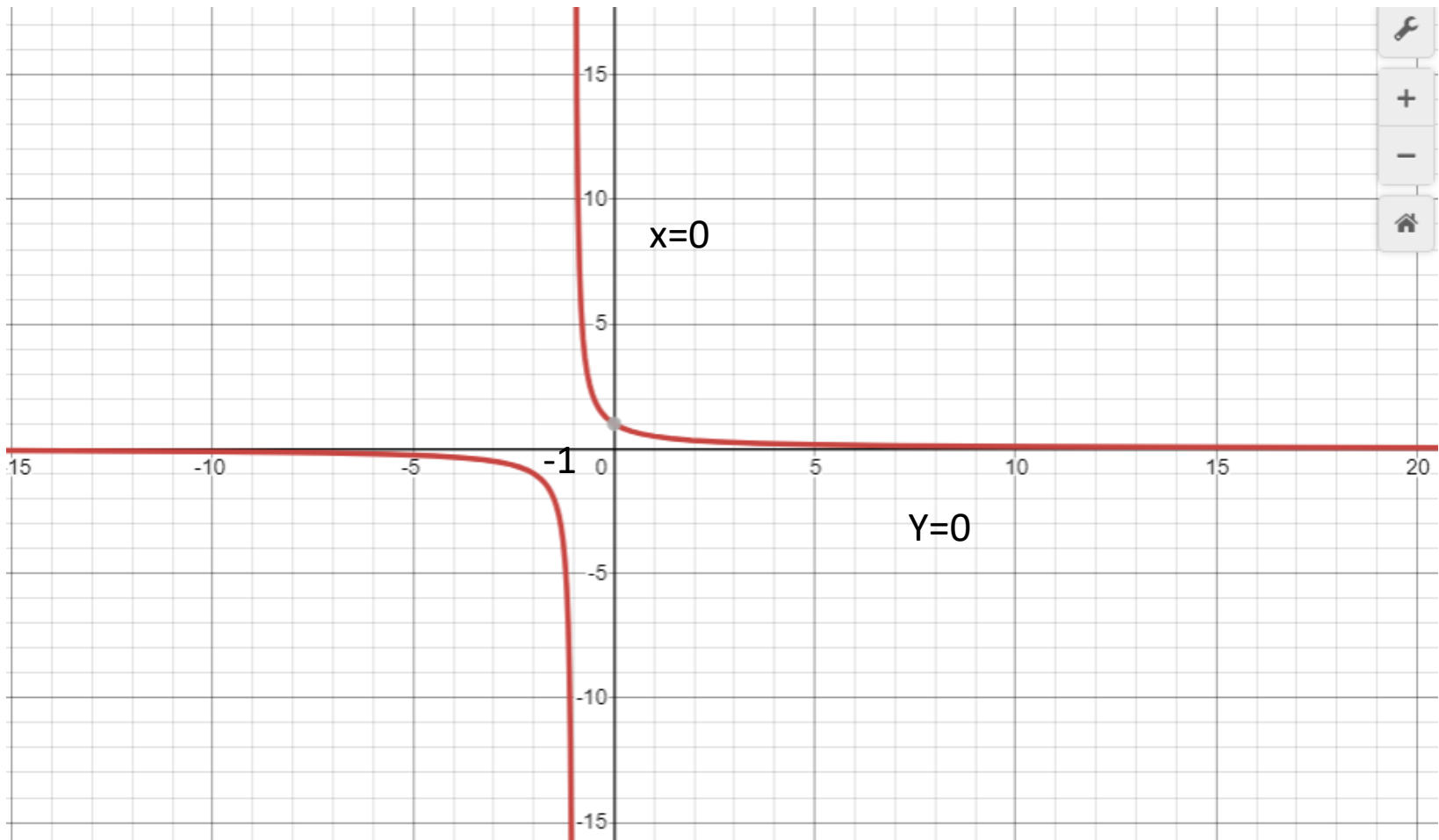
Sketch the graph of $f(x)=y = \frac{1}{x-3}$, also find their domain and range.



Finding domain and range $y = \frac{1}{x-3}$.

- Since f is undefined if $x - 3 = 0$
 $\Rightarrow x = 3$ so domain is $x \neq 3$.
- Here $y = \frac{1}{x-3} \Rightarrow xy - 3y = 1$
- $\Rightarrow xy = 1 + 3y \Rightarrow x = \frac{1+3y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{3\}$ and Range is : $R - \{0\}$

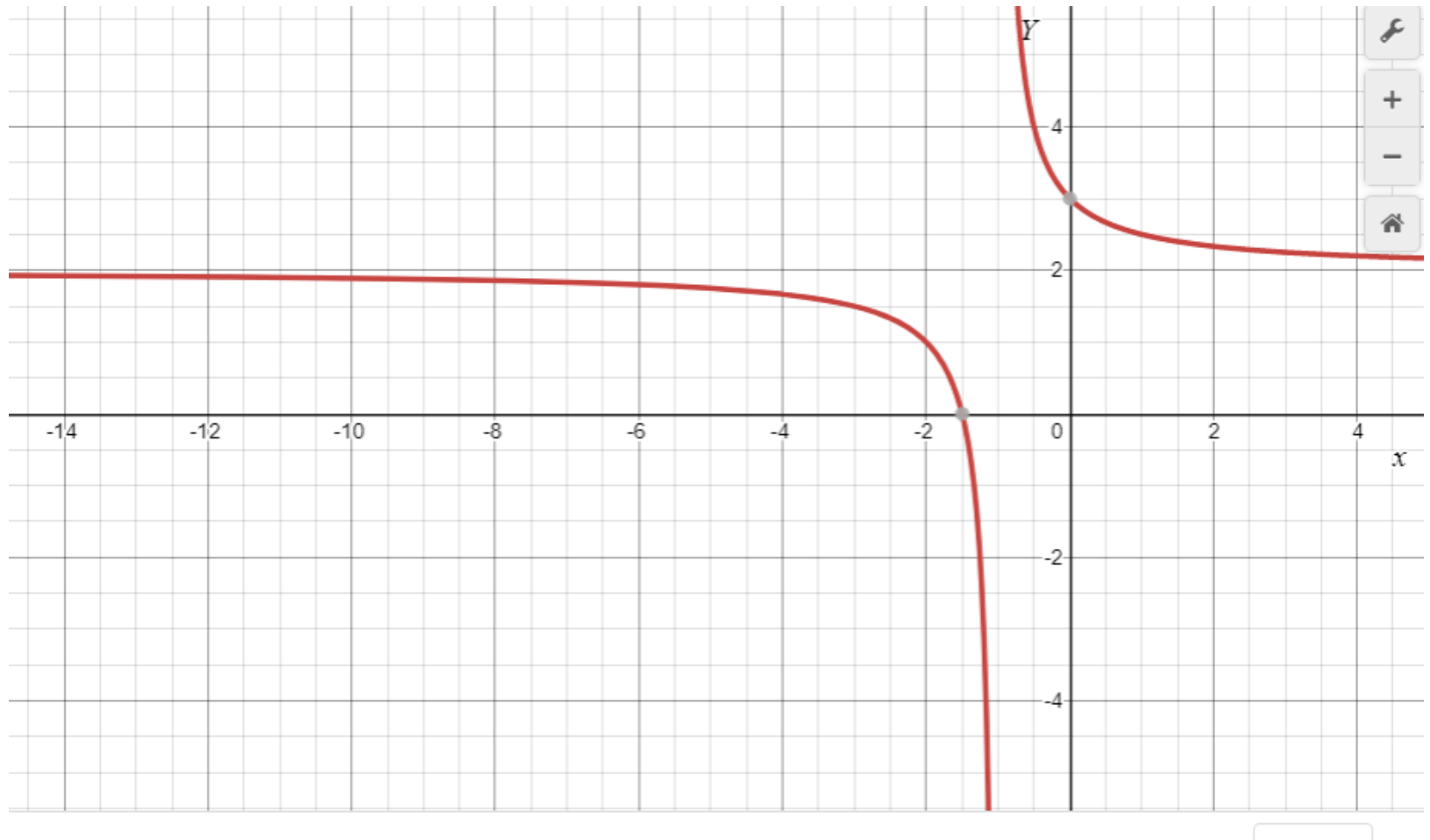
Sketch the graph of $f(x)=y = \frac{1}{1+x}$, also find their domain and range.



Finding domain and range $y = \frac{1}{1+x}$.

- Since f is undefined if $x + 1 = 0$
 $\Rightarrow x = -1$ so domain is $x \neq -1$.
- Here $y = \frac{1}{x+1} \Rightarrow xy + y = 1$
- $\Rightarrow xy = 1 - y \Rightarrow x = \frac{1-y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{-1\}$ and Range is : $R - \{0\}$

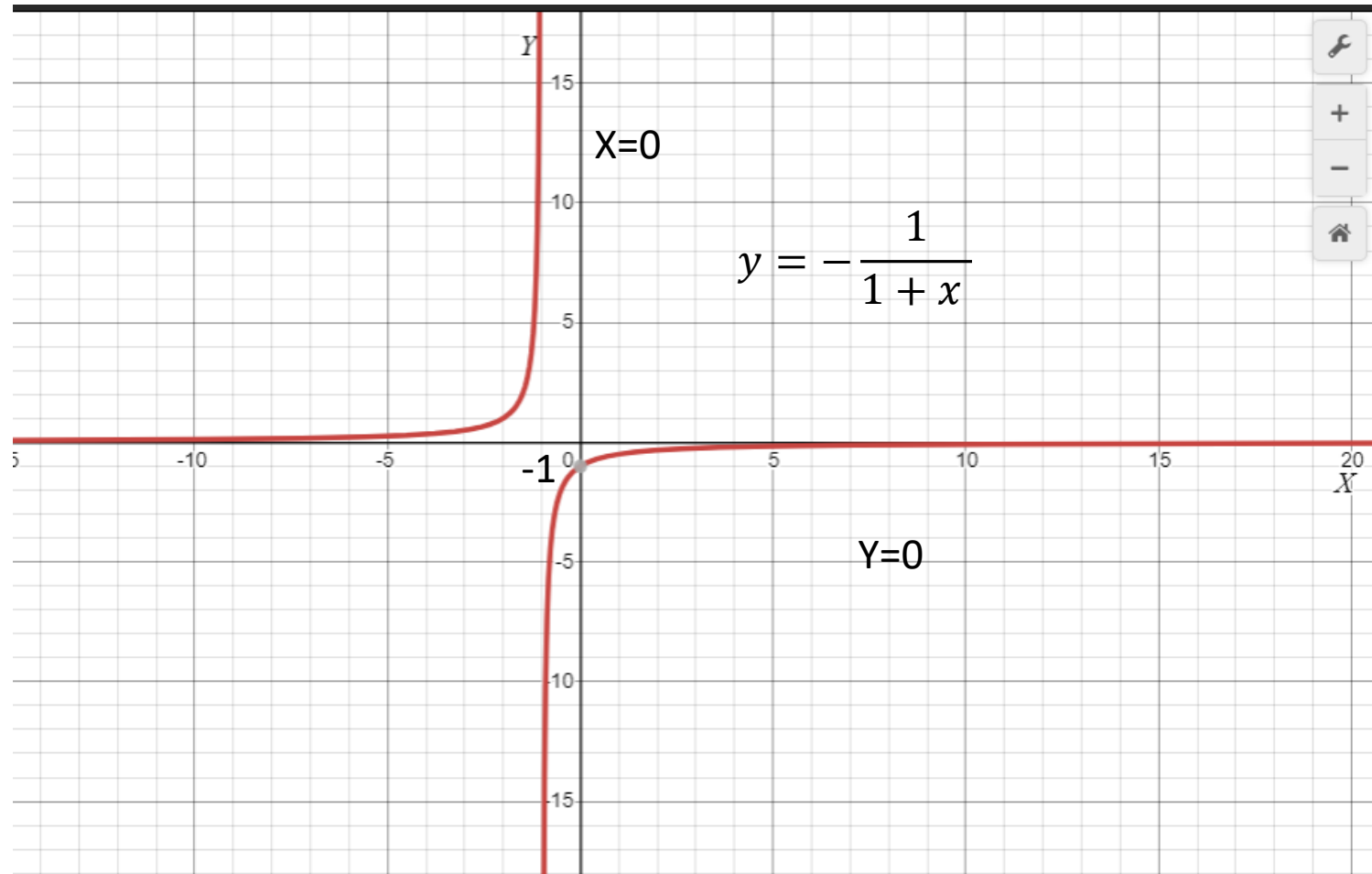
Sketch the graph of $f(x)=y = 2 + \frac{1}{1+x}$, also find their domain and range.



Finding domain and range $y = 2 + \frac{1}{1+x}$.

- Since f is undefined if $x + 1 = 0$
 $\Rightarrow x = -1$ so domain is $x \neq -1$.
- Here $y = 2 + \frac{1}{x+1} \Rightarrow y - 2 = \frac{1}{x+1}$
 $\Rightarrow -2x + xy + y - 2 = 1$
 $\Rightarrow -2x + xy = 1 - y + 2 \Rightarrow x(y - 2) = -y + 3$
 $\Rightarrow x = \frac{3-y}{y-2}, y \neq 2$
- So range is $y \neq 2$
- Or
- Domain is : $R - \{-1\}$ and Range is : $R - \{2\}$

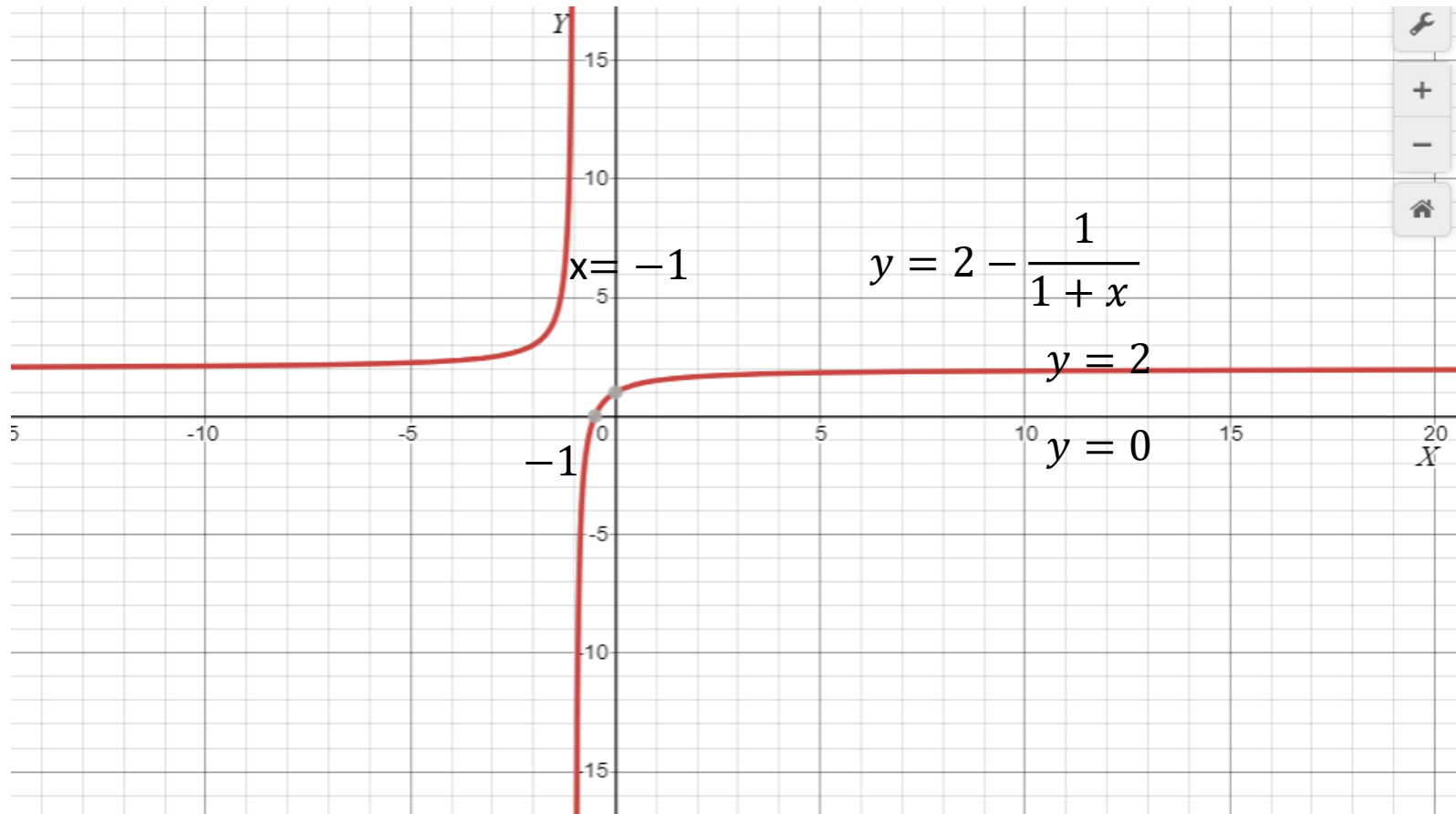
Sketch the graph of $f(x)=y = -\frac{1}{1-x}$, also find their domain and range.



Finding domain and range $y = -\frac{1}{1+x}$.

- Since f is undefined if $x + 1 = 0$
 $\Rightarrow x = -1$ so domain is $x \neq -1$.
- Here $y = -\frac{1}{x+1} \Rightarrow xy + y = -1$
- $\Rightarrow xy = -1 - y \Rightarrow x = \frac{-1-y}{y}, y \neq 0$.
- So range is $y \neq 0$
- Or
- Domain is : $R - \{-1\}$ and Range is : $R - \{0\}$

Sketch the graph of $f(x)=y = 2 - \frac{1}{1+x}$, also find their domain and range.



Finding domain and range $y = 2 - \frac{1}{1+x}$.

- Since f is undefined if $x + 1 = 0$

$\Rightarrow x = -1$ so domain is $x \neq -1$.

- Here $y = 2 - \frac{1}{x+1} \Rightarrow 2 - y = \frac{1}{x+1}$
 $\Rightarrow 2x - xy + 2 - y = 1$

$$\Rightarrow 2x - xy = 1 + y - 2 \Rightarrow x(2 - y) = y - 1$$

$$\Rightarrow x = \frac{y-1}{2-y}, y \neq 2$$

- So range is $y \neq 2$

- Or

- Domain is : $R - \{-1\}$ and Range is : $R - \{2\}$

0.2.1 DEFINITION Given functions f and g , we define

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$(f/g)(x) = f(x)/g(x)$$

For the functions $f + g$, $f - g$, and fg we define the domain to be the intersection of the domains of f and g , and for the function f/g we define the domain to be the intersection of the domains of f and g but with the points where $g(x) = 0$ excluded (to avoid division by zero).

0.2

NEW FUNCTIONS FROM OLD

► **Example 1** Let

$$f(x) = 1 + \sqrt{x-2} \quad \text{and} \quad g(x) = x - 3$$

Find the domains and formulas for the functions $f + g$, $f - g$, fg , f/g , and $7f$.

Solution. First, we will find the formulas and then the domains. The formulas are

$$(f + g)(x) = f(x) + g(x) = (1 + \sqrt{x-2}) + (x - 3) = x - 2 + \sqrt{x-2} \quad (2)$$

$$(f - g)(x) = f(x) - g(x) = (1 + \sqrt{x-2}) - (x - 3) = 4 - x + \sqrt{x-2} \quad (3)$$

$$(fg)(x) = f(x)g(x) = (1 + \sqrt{x-2})(x - 3) \quad (4)$$

$$(f/g)(x) = f(x)/g(x) = \frac{1 + \sqrt{x-2}}{x-3} \quad (5)$$

$$(7f)(x) = 7f(x) = 7 + 7\sqrt{x-2} \quad (6)$$

The domains of f and g are $[2, +\infty)$ and $(-\infty, +\infty)$, respectively (their natural domains). Thus, it follows from Definition 0.2.1 that the domains of $f + g$, $f - g$, and fg are the intersection of these two domains, namely,

$$[2, +\infty) \cap (-\infty, +\infty) = [2, +\infty) \quad (7)$$

Moreover, since $g(x) = 0$ if $x = 3$, the domain of f/g is (7) with $x = 3$ removed, namely,

$$[2, 3) \cup (3, +\infty)$$

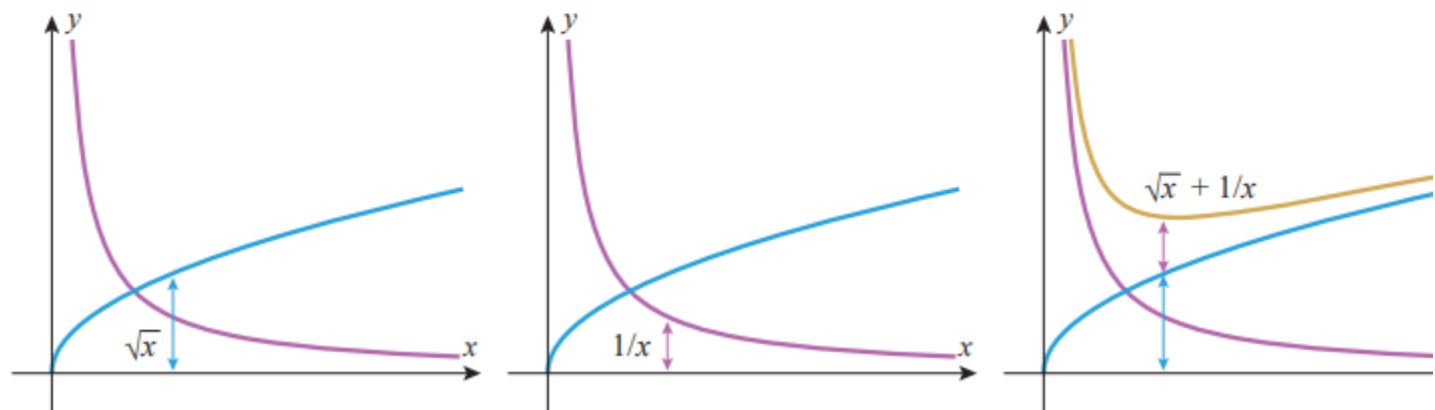
Finally, the domain of $7f$ is the same as the domain of f . ◀

0.2

NEW FUNCTIONS FROM OLD

► **Example 6** Referring to Figure 0.1.4 for the graphs of $y = \sqrt{x}$ and $y = 1/x$, make a sketch that shows the general shape of the graph of $y = \sqrt{x} + 1/x$ for $x \geq 0$.

Solution. To add the corresponding y -values of $y = \sqrt{x}$ and $y = 1/x$ graphically, just imagine them to be “stacked” on top of one another. This yields the sketch in Figure 0.2.2



0.2

NEW FUNCTIONS FROM OLD

► **Example 2** Show that if $f(x) = \sqrt{x}$, $g(x) = \sqrt{x}$, and $h(x) = x$, then the domain of fg is not the same as the natural domain of h .

Solution. The natural domain of $h(x) = x$ is $(-\infty, +\infty)$. Note that

$$(fg)(x) = \sqrt{x}\sqrt{x} = x = h(x)$$

on the domain of fg . The domains of both f and g are $[0, +\infty)$, so the domain of fg is

$$[0, +\infty) \cap [0, +\infty) = [0, +\infty)$$

■ COMPOSITION OF FUNCTIONS

■ COMPOSITION OF FUNCTIONS

We now consider an operation on functions, called *composition*, which has no direct analog in ordinary arithmetic. Informally stated, the operation of composition is performed by substituting some function for the independent variable of another function. For example, suppose that

$$f(x) = x^2 \quad \text{and} \quad g(x) = x + 1$$

If we substitute $g(x)$ for x in the formula for f , we obtain a new function

$$f(g(x)) = (g(x))^2 = (x + 1)^2$$

which we denote by $f \circ g$. Thus,

$$(f \circ g)(x) = f(g(x)) = (g(x))^2 = (x + 1)^2$$

In general, we make the following definition.

0.2.2 DEFINITION Given functions f and g , the *composition* of f with g , denoted by $f \circ g$, is the function defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is defined to consist of all x in the domain of g for which $g(x)$ is in the domain of f .

► **Example 3** Let $f(x) = x^2 + 3$ and $g(x) = \sqrt{x}$. Find

$$(a) (f \circ g)(x) \quad (b) (g \circ f)(x)$$

Solution (a). The formula for $f(g(x))$ is

$$f(g(x)) = [g(x)]^2 + 3 = (\sqrt{x})^2 + 3 = x + 3$$

Since the domain of g is $[0, +\infty)$ and the domain of f is $(-\infty, +\infty)$, the domain of $f \circ g$ consists of all x in $[0, +\infty)$ such that $g(x) = \sqrt{x}$ lies in $(-\infty, +\infty)$; thus, the domain of $f \circ g$ is $[0, +\infty)$. Therefore,

$$(f \circ g)(x) = x + 3, \quad x \geq 0$$

Solution (b). The formula for $g(f(x))$ is

$$g(f(x)) = \sqrt{f(x)} = \sqrt{x^2 + 3}$$

Since the domain of f is $(-\infty, +\infty)$ and the domain of g is $[0, +\infty)$, the domain of $g \circ f$ consists of all x in $(-\infty, +\infty)$ such that $f(x) = x^2 + 3$ lies in $[0, +\infty)$. Thus, the domain of $g \circ f$ is $(-\infty, +\infty)$. Therefore,

$$(g \circ f)(x) = \sqrt{x^2 + 3}$$

There is no need to indicate that the domain is $(-\infty, +\infty)$, since this is the natural domain of $\sqrt{x^2 + 3}$. ◀

Compositions can also be defined for three or more functions; for example, $(f \circ g \circ h)(x)$ is computed as

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

In other words, first find $h(x)$, then find $g(h(x))$, and then find $f(g(h(x)))$.

► **Example 4** Find $(f \circ g \circ h)(x)$ if

$$f(x) = \sqrt{x}, \quad g(x) = 1/x, \quad h(x) = x^3$$

Solution.

$$(f \circ g \circ h)(x) = f(g(h(x))) = f(g(x^3)) = f(1/x^3) = \sqrt{1/x^3} = 1/x^{3/2} \quad \blacktriangleleft$$

■ EXPRESSING A FUNCTION AS A COMPOSITION

Many problems in mathematics are solved by “decomposing” functions into compositions of simpler functions. For example, consider the function h given by

$$h(x) = (x + 1)^2$$

To evaluate $h(x)$ for a given value of x , we would first compute $x + 1$ and then square the result. These two operations are performed by the functions

$$g(x) = x + 1 \quad \text{and} \quad f(x) = x^2$$

We can express h in terms of f and g by writing

$$h(x) = (x + 1)^2 = [g(x)]^2 = f(g(x))$$

so we have succeeded in expressing h as the composition $h = f \circ g$.

The thought process in this example suggests a general procedure for decomposing a function h into a composition $h = f \circ g$:

- Think about how you would evaluate $h(x)$ for a specific value of x , trying to break the evaluation into two steps performed in succession.
- The first operation in the evaluation will determine a function g and the second a function f .
- The formula for h can then be written as $h(x) = f(g(x))$.

► **Example 5** Express $\sin(x^3)$ as a composition of two functions.

Solution. To evaluate $\sin(x^3)$, we would first compute x^3 and then take the sine, so $g(x) = x^3$ is the inside function and $f(x) = \sin x$ the outside function. Therefore,

$$\sin(x^3) = f(g(x)) \quad \boxed{g(x) = x^3 \text{ and } f(x) = \sin x} \quad \blacktriangleleft$$

Table 0.2.1 gives some more examples of decomposing functions into compositions.

Table 0.2.1
COMPOSING FUNCTIONS

FUNCTION	$g(x)$ INSIDE	$f(x)$ OUTSIDE	COMPOSITION
$(x^2 + 1)^{10}$	$x^2 + 1$	x^{10}	$(x^2 + 1)^{10} = f(g(x))$
$\sin^3 x$	$\sin x$	x^3	$\sin^3 x = f(g(x))$
$\tan(x^5)$	x^5	$\tan x$	$\tan(x^5) = f(g(x))$
$\sqrt{4 - 3x}$	$4 - 3x$	$\sqrt{x}$	$\sqrt{4 - 3x} = f(g(x))$
$8 + \sqrt{x}$	$\sqrt{x}$	$8 + x$	$8 + \sqrt{x} = f(g(x))$
$\frac{1}{x+1}$	$x+1$	$\frac{1}{x}$	$\frac{1}{x+1} = f(g(x))$

REMARK There is always more than one way to express a function as a composition. For example, here are two ways to express $(x^2 + 1)^{10}$ as a composition that differ from that in Table 0.2.1:

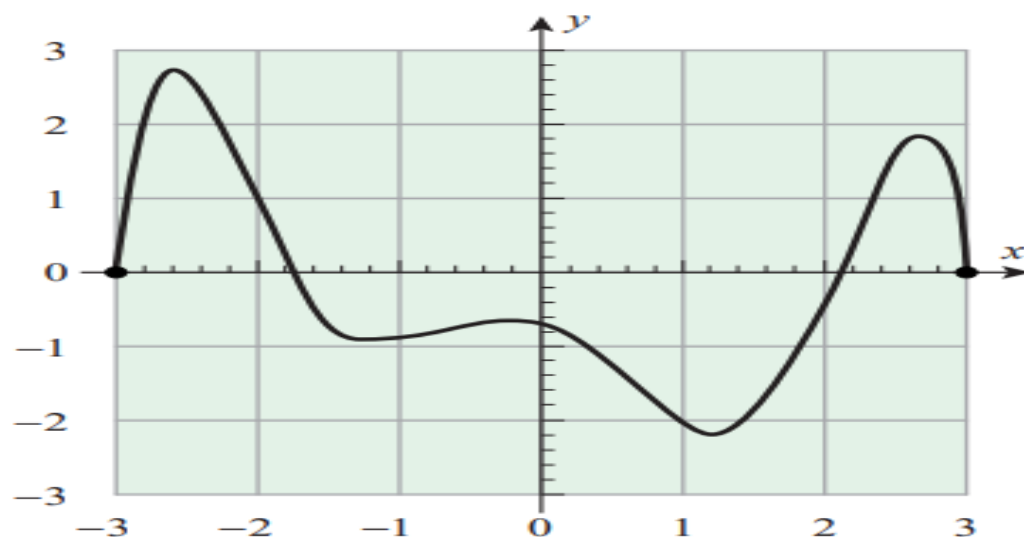
$$(x^2 + 1)^{10} = [(x^2 + 1)^2]^5 = f(g(x))$$

$$g(x) = (x^2 + 1)^2 \text{ and } f(x) = x^5$$

$$(x^2 + 1)^{10} = [(x^2 + 1)^3]^{10/3} = f(g(x))$$

$$g(x) = (x^2 + 1)^3 \text{ and } f(x) = x^{10/3}$$

1. Use the accompanying graph to answer the following questions, making reasonable approximations where needed.
- (a) For what values of x is $y = 1$?
 - (b) For what values of x is $y = 3$?
 - (c) For what values of y is $x = 3$?
 - (d) For what values of x is $y \leq 0$?
 - (e) What are the maximum and minimum values of y and for what values of x do they occur?

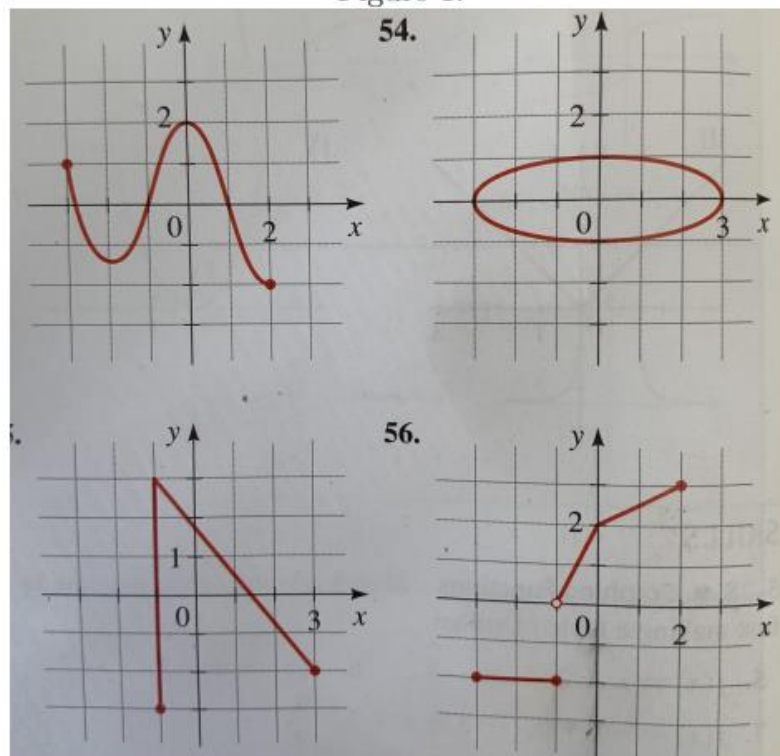


◀ **Figure Ex-1**

1. (a) $-2.9, -2.0, 2.35, 2.9$ (b) None (c) $y = 0$ (d) $-1.75 \leq x \leq 2.15, x = -3, x = 3$
- (e) $y_{\max} = 2.8$ at $x = -2.6$; $y_{\min} = -2.2$ at $x = 1.2$

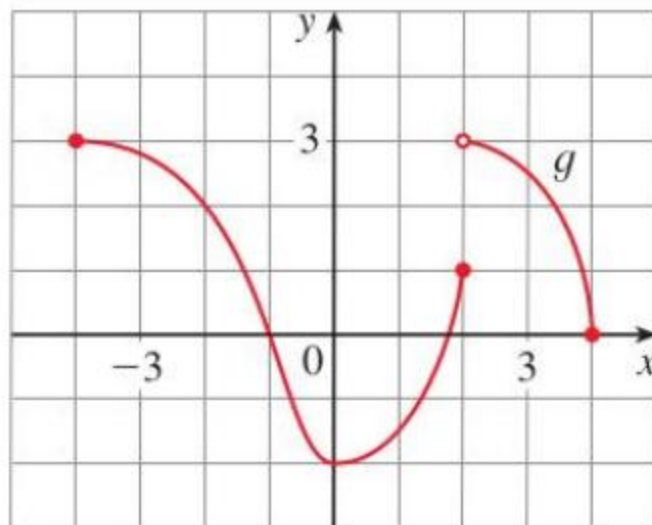
1. Determine whether the curve is the graph (Figure 1) of a function of x . If not, sketch domain and range of the function along with the corresponding function.

Figure 1:



2. The graph (Figure 2) of a function g is given.

Figure 2:



- (a) State the values of x where $g(x)$ are -2 , 3 , 1 and 0 .
- (b) On what interval(s) is $g(x) \leq 0$.
- (c) Sketch the domain and the range of the function g .
- (d) On what interval(s) is g decreasing and decreasing?

■ STRETCHES AND COMPRESSIONS

Horizontal And Vertical Graph Stretches And Compressions

- $y = c f(x)$, vertical stretch, factor of c .
- $y = (1/c)f(x)$, compress vertically, factor of c .
- $y = f(cx)$, compress horizontally, factor of c .
- $y = f(x/c)$, stretch horizontally, factor of c .
- $y = -f(x)$, reflect at x -axis.
- $y = f(-x)$, reflect at y -axis.

Exercise

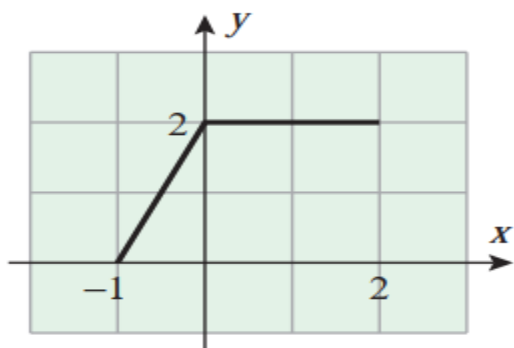
1. The graph of a function f is shown in the accompanying figure. Sketch the graphs of the following equations.

(a) $y = f(x) - 1$

(b) $y = f(x - 1)$

(c) $y = \frac{1}{2}f(x)$

(d) $y = f\left(-\frac{1}{2}x\right)$



◀ **Figure Ex-1**

2. Use the graph in Exercise 1 to sketch the graphs of the following equations.

(a) $y = -f(-x)$

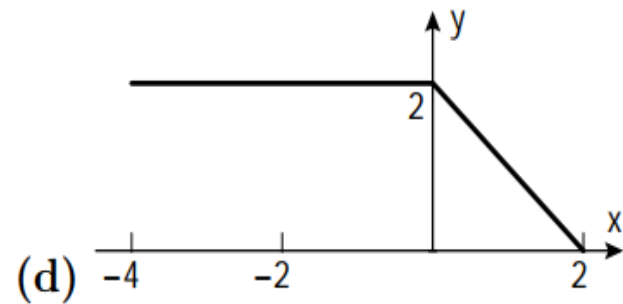
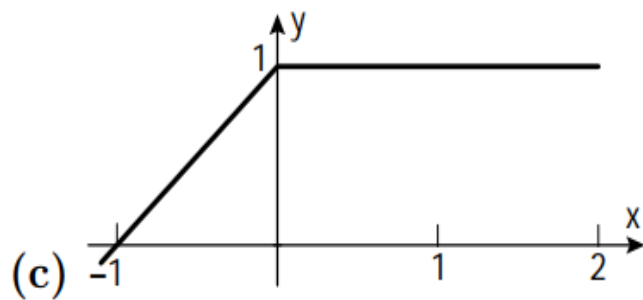
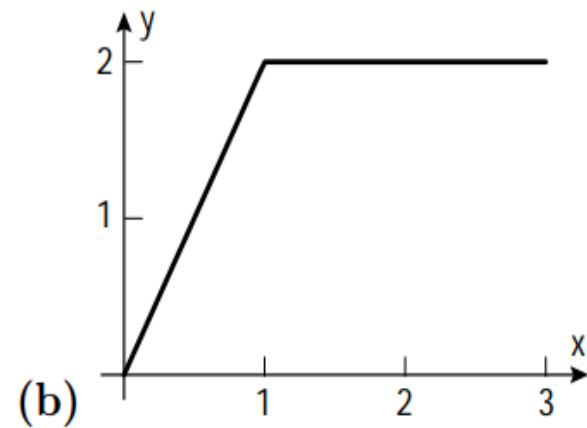
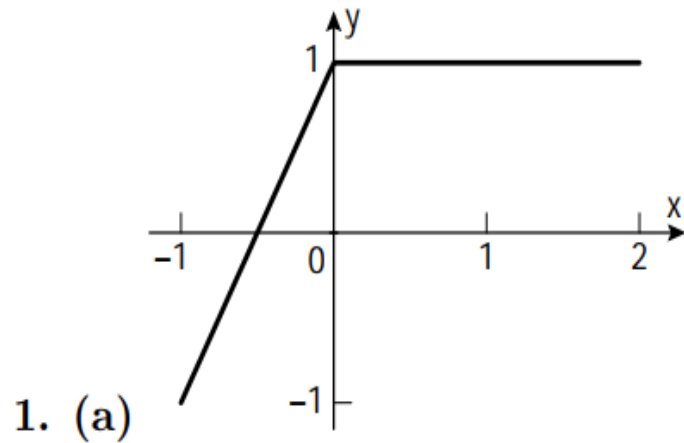
(b) $y = f(2 - x)$

(c) $y = 1 - f(2 - x)$

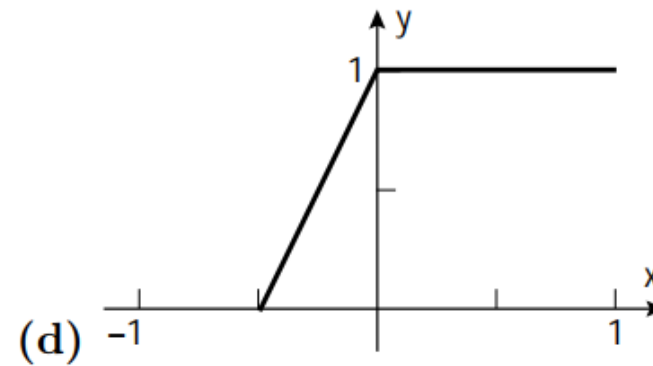
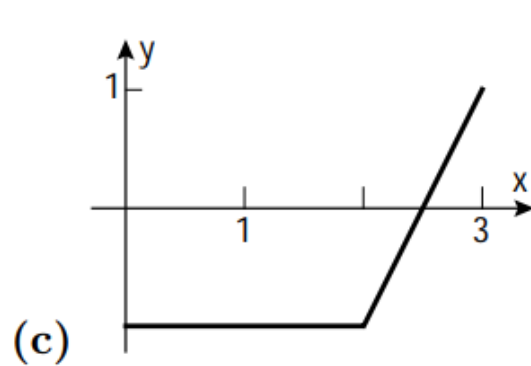
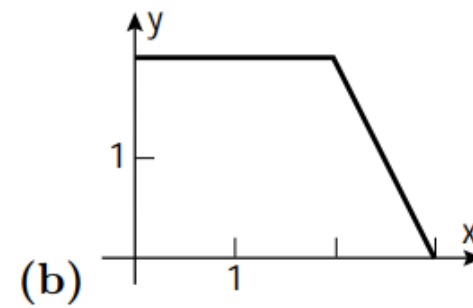
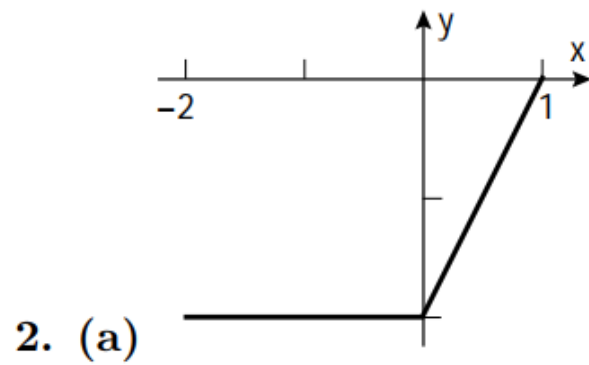
(d) $y = \frac{1}{2}f(2x)$

solution

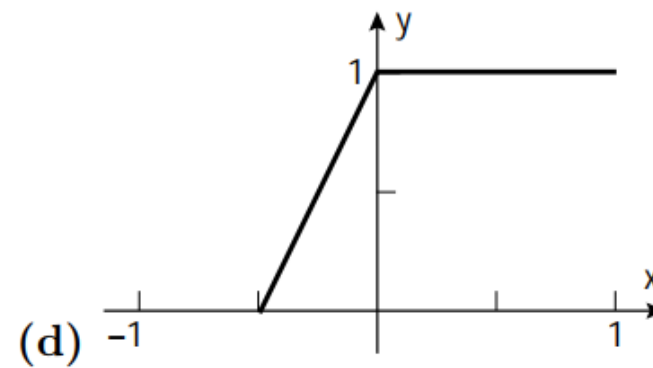
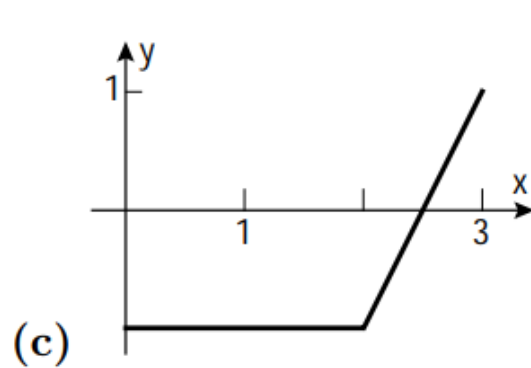
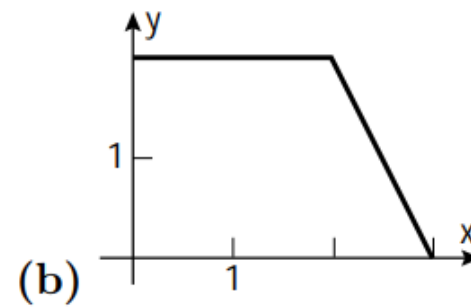
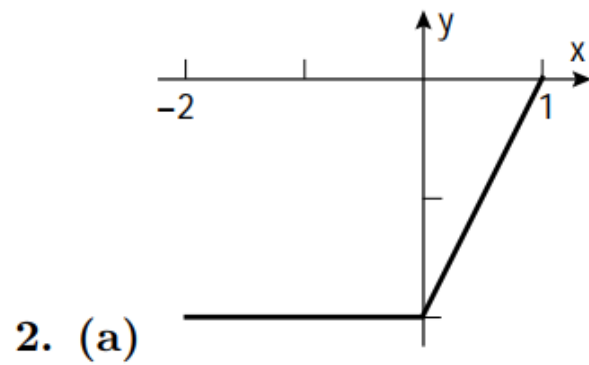
- 1.



Solution



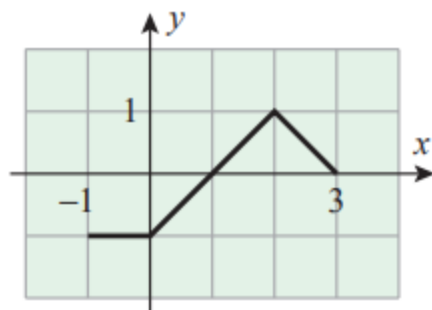
Solution



3. The graph of a function f is shown in the accompanying figure. Sketch the graphs of the following equations.

(a) $y = f(x + 1)$

(b) $y = f(2x)$

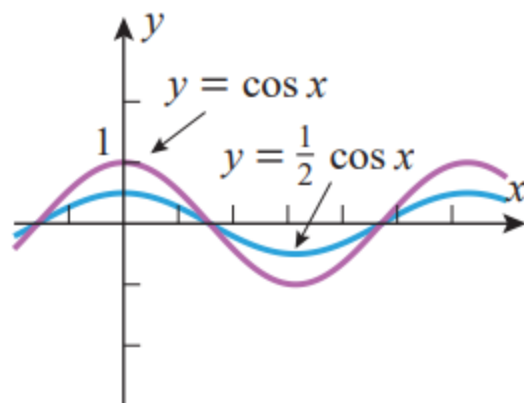
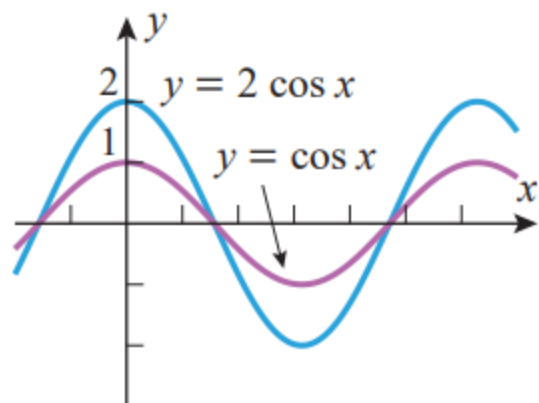


◀ **Figure Ex-3**

(c) $y = |f(x)|$

(d) $y = 1 - |f(x)|$

■ STRETCHES AND COMPRESSIONS



■ STRETCHES AND COMPRESSIONS

